



Zero forcing of generalized hierarchical products

Sarah E. Anderson, Brenda K. Kroschel*

Department of Mathematics, University of St. Thomas, St. Paul, MN 55105, USA

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ABSTRACT

Zero forcing is a process that passes information along a network. In particular, a vertex that has information may pass that information to one of its neighbors only if that neighbor is its only neighbor that does not have that information. If the information is given to an initial set of vertices that eventually passes the information to the entire network, then that initial set is called a zero forcing set. The zero forcing number of a graph is the size of a minimum zero forcing set of that graph. In this paper, bounds on the zero forcing number of generalized hierarchical products, which are a generalization of the Cartesian product, are provided. In particular, the zero forcing number is determined exactly for certain hierarchical products by constructing zero forcing sets in the product that utilize knowledge about forcing chains for each factor of the product.

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1. Introduction

Zero forcing is a combinatorial process on a graph by which information is passed between neighboring vertices following a given set of rules. The concept of zero forcing has been studied in a number of branches of science and mathematics. Applications include inverse eigenvalue problems, social networking, graph search algorithms, logic circuits, and quantum control [6,7,9,10,13]. Zero forcing was originally developed in the combinatorial matrix community to provide a bound for the minimum rank of a symmetric matrix associated with a graph [1].

In this paper, simple, undirected, connected graphs are considered. The usual *zero forcing* process passes information along the graph using the following *forcing rule* (sometimes called the color-change rule, or filling rule): if a vertex v_n is filled, and has only one neighbor v_m that is unfilled, then v_n forces v_m to be filled, and is denoted $v_n \rightarrow v_m$. Given an initial set of filled vertices, the final set of vertices that are filled after successively applying the forcing rule until no more changes are possible is called the *derived set* [1], and this set is unique. The goal is to find a minimum initial set of vertices that passes information along to the entire graph; that is, a minimum initial set of vertices whose derived set is the entire graph. An initial set of filled vertices that results in a derived set that is the entire graph is called a *zero forcing set*. The filled vertices on the left-hand graph of Fig. 1.1 form a zero forcing set, while those on the right-hand graph do not. The *zero forcing number*, $Z(G)$, of a graph G is the cardinality of a minimum zero forcing set.

This paper studies the zero forcing number of the generalized hierarchical product. The generalized hierarchical product of graphs was first introduced in 2009 in [5]. The generalized hierarchical product, which is a generalization of the hierarchical product and the Cartesian product, can be used to model scale-free networks [4]. There are many real-world examples of scale-free networks, including the World Wide Web, collaboration networks, and metabolic networks [12,15]. In this paper, the “generalized hierarchical product” will be referred to as the “hierarchical product”.

* Corresponding author.

E-mail addresses: ande1298@stthomas.edu (S.E. Anderson), bkkroschel@stthomas.edu (B.K. Kroschel).

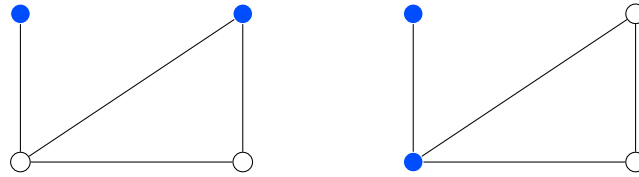


Fig. 1.1. The filled vertices on the left are a minimum zero forcing set. The filled vertices on the right are not a zero forcing set.

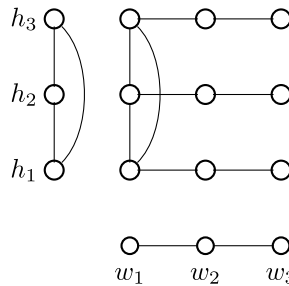


Fig. 1.2. $G = P_3(U) \square C_3$, $|U| = \{w_1\}$.

Given graphs W and H and vertex subset $U \subseteq V(W)$, the *hierarchical product*, denoted $G = W(U) \square H$, is the graph with vertex set $V(W) \times V(H)$ in which two vertices (x_1, y_1) and (x_2, y_2) are adjacent in $G = W(U) \square H$ if either $x_1 \in U$, $x_1 = x_2$ and $y_1 y_2$ is an edge in H , or $y_1 = y_2$ and $x_1 x_2$ is an edge in W . Note that by convention, the copies of W are referred to as rows and those of H are referred to as columns. The root set, U , determines the edge set for the columns of the graph. That is, $W(U) \square H$ contains $|U|$ copies of H as well as $|V(H)|$ copies of W . Fig. 1.2 depicts a hierarchical graph product of a path and cycle with a root set $U = \{w_1\}$. Notice that there are three copies of P_3 across the rows as well as one copy of C_3 along the vertex of the root set. We refer to forcing along a copy of W as horizontal forcing. Similarly, vertical forcing is along a copy of H .

The *Cartesian product* of two graphs W and H , denoted $G = W \square H$, is a special case of the hierarchical product where the root set is all of the vertices of W . In [1], an upper bound on the zero forcing number of the Cartesian product of any two arbitrary graphs was given, which is generalized in Lemma 2.4. In addition, the zero forcing number of the Cartesian product with particular factors, such as $P_s \square P_t$, $C_s \square P_t$, $K_s \square K_t$, and $C_s \square K_t$, was determined exactly in [1].

In this paper, the zero forcing number of hierarchical products for various root sets is determined. In Section 2, we provide general lower and upper bounds on $Z(W(U) \square H)$ for graphs W and H with any root set $U \subseteq V(W)$. In addition, $Z(W(U) \square H)$ is found exactly for particular root sets. In Section 3, an upper bound on $Z(W(U) \square H)$ for graphs W and H with any root set $U \subseteq V(W)$ is determined by using forcing chains from each factor. These bounds are used in Section 4 to find the zero forcing number of certain hierarchical products where at least one factor has a relatively large zero forcing number.

2. General bounds

In this section, general bounds on $Z(W(U) \square H)$ for any graphs W and H with any root set $U \subseteq V(W)$ are constructed. First, some notation that is used throughout the section is established. Let S be a subset of $V(W(U) \square H)$. Then $S_{PW} := \{w \mid (w, h) \in S\}$ is the *projection of S onto W* . In addition, let $S_{PH} := \{h \mid (w, h) \in S\}$ be the *projection of S onto H* .

Lemma 2.1. *For any connected graphs W and H with root set $U \subseteq V(W)$, $\max\{Z(W), Z(H)\} \leq Z(W(U) \square H)$.*

Proof. Let S be a minimum zero forcing set of $W(U) \square H$.

First, show that $Z(W(U) \square H) \geq Z(W)$. Let S_{PW} be the projection of S onto W . Note that $|S| \geq |S_{PW}|$. If S_{PW} is a zero forcing set of W , then $Z(W(U) \square H) = |S| \geq |S_{PW}| \geq Z(W)$ which is greater than $\max\{Z(W), Z(H)\}$.

Now, assume for the sake of contradiction that S_{PW} is not a zero forcing set of W . Let $S_1 = S_{PW} \times V(H)$. That is, S_1 consists of the vertices corresponding to S_{PW} in each copy of W . Note that $S \subseteq S_1$. Thus, S_1 is a zero forcing set of $W(U) \square H$, and S_1 will only force vertices horizontally along copies of W . So it must be the case that S_{PW} is a zero forcing set of each copy of W in $W(U) \square H$, which is a contradiction.

Next, show that $Z(W(U) \square H) \geq Z(H)$. Let S_{PH} be the projection of S onto H . As before, if S_{PH} is a zero forcing set of H , then the inequality holds. So, we may assume for the sake of contradiction that S_{PH} is not a zero forcing set of H . Let $S_2 = (U \times S_{PH}) \cup \{(w, h) \mid w \notin U \text{ and } h \in V(H)\}$. Hence, S_2 consists of the vertices corresponding to S_{PH} in each copy of H

as well as all of the remaining vertices in the columns not associated with U . Note $S \subseteq S_2$. Thus, S_2 is a zero forcing set of $W(U) \cap H$, and S_2 will only force vertices vertically along copies of H . Therefore, S_{PH} is a zero forcing set of each copy of H in $W(U) \cap H$, which is a contradiction. $\square$

For root sets U that are small relative to $|V(W)|$, [Theorem 2.2](#) provides an improved lower bound to [Lemma 2.1](#).

Theorem 2.2. For any connected graphs W and H with root set $U \subseteq V(W)$,

$$Z(W)|V(H)| - |V(H)||U| + Z(H)|U| \leq Z(W(U) \cap H).$$

Proof. Let S be a minimum zero forcing set of $W(U) \cap H$. Note that for each copy of W , the maximum number of vertices that can be forced horizontally is $|V(W)| - Z(W)$, and there are $|V(H)|$ copies of W . Similarly, for each copy of H , the maximum number of vertices that can be forced vertically is $|V(H)| - Z(H)$, and there are $|U|$ copies of H . Since all vertices are forced by the end of the forcing process, $|S| + (|V(W)| - Z(W))|V(H)| + (|V(H)| - Z(H))|U| \geq |V(W)||V(H)|$. Thus,

$$\begin{aligned} |S| &\geq |V(W)||V(H)| - (|V(W)| - Z(W))|V(H)| - (|V(H)| - Z(H))|U| \\ &= Z(W)|V(H)| - |V(H)||U| + Z(H)|U|. \quad \square \end{aligned}$$

For certain root sets, [Theorem 2.3](#) is an improvement of the lower bounds given in [Lemma 2.1](#) and [Theorem 2.2](#). In fact, if S' is a smallest zero forcing set of W that contains the root set U and has size $|S'| = Z(W) + |U|$, then [Theorem 2.3](#) states $Z(W)|V(H)| \leq Z(W(U) \cap H)$.

Theorem 2.3. Let W and H be connected graphs with root set $U \subseteq V(W)$. Let S' be any smallest zero forcing set of W such that $U \subseteq S'$. Then, every zero forcing set of $W(U) \cap H$ contains at least $|S'| - |U|$ vertices in each copy of W and

$$(|S'| - |U|)|V(H)| \leq Z(W(U) \cap H).$$

Proof. Note that if $|S'| = |U|$, then $0 \leq Z(W(U) \cap H)$. So, we may assume $|S'| > |U|$. For the sake of contradiction, assume there exists a zero forcing set T of $W(U) \cap H$ such that T does not contain $|S'| - |U|$ vertices in at least one copy of W . That is, assume T contains $t < |S'| - |U|$ vertices in one such copy of W . Note that the maximum number of vertices in this copy of W that could be filled by vertical forcing is $|U|$. Thus, in the best case scenario, for this copy of W , $|U|$ vertices are forced vertically and t vertices are in the zero forcing set T . Since T is a zero forcing set of $W(U) \cap H$, the remaining vertices in this copy of W are forced horizontally. Therefore, there exists a zero forcing set of W that contains U of size $|U| + t < |U| + |S'| - |U| = |S'|$. This is a contradiction since S' is the smallest zero forcing set of W that contains U . Since every zero forcing set of $W(U) \cap H$ contains at least $|S'| - |U|$ vertices in each copy of W , and there are $|V(H)|$ copies of W , $(|S'| - |U|)|V(H)| \leq Z(W(U) \cap H)$. $\square$

An upper bound for the zero forcing number of the hierarchical product of two graphs was given in [\[14\]](#). Note that this result is a generalization of an observation given in [\[1\]](#), which states $Z(W \square H) \leq \min\{Z(W)|V(H)|, Z(H)|V(W)|\}$.

Lemma 2.4 ([\[14\]](#)). Let W and H be any connected graphs with root set $U \subseteq V(W)$. Then

$$Z(W(U) \cap H) \leq \min\{Z(W)|V(H)|, Z(H)|U| + (|V(W)| - |U|)|V(H)|\}.$$

The following corollary follows from [Theorem 2.3](#) and [Lemma 2.4](#).

Corollary 2.5. Let W and H be any connected graphs with root set $U \subseteq V(W)$.

- (1) If U is a subset of a minimum zero forcing set of W , then $(Z(W) - |U|)|V(H)| \leq Z(W(U) \cap H) \leq Z(W)|V(H)|$.
- (2) If S' is any smallest zero forcing set of W that contains U and has size $|S'| = Z(W) + |U|$, then $Z(W(U) \cap H) = Z(W)|V(H)|$.
- (3) If $U = \{u\}$ and u is not contained in any minimum zero forcing set of W , then $Z(W(U) \cap H) = Z(W)|V(H)|$.

Example 2.6. Consider $W = P_n$, where P_n denotes the path on n vertices. Assume U is not a zero forcing set of P_n . Since $Z(P_n) = 1$, the smallest zero forcing set of P_n that contains U has size $|U| + 1$. Thus, by [Corollary 2.5 \(2\)](#), $Z(P_n(U) \cap H) = |V(H)|$, in which H is any connected graph.

Theorem 2.7. Let W and H be any connected graphs. Let $U \subseteq V(W)$ be a root set such that for every $u \in U$, u is adjacent to two or more pendant vertices of W . Then,

$$Z(W(U) \cap H) = Z(W)|V(H)|.$$

Proof. Let S be any smallest zero forcing set of W that contains U . We claim $|S| = Z(W) + |U|$. For the sake of contradiction, assume $|S| < Z(W) + |U|$. Let $u \in U$. Then, u is adjacent to two or more pendant vertices of W . Consider $S' = S \setminus \{u\}$. It will be shown that S' is a zero forcing set of W .

Case 1: Assume u does not force any vertices in W . Recall that u is adjacent to two or more pendant vertices. Since S is a zero forcing set of W , those pendant vertices must either be in S or be forced by u . Since u does not force any vertices in W , all pendant vertices adjacent to u must be in S . So, one of those pendant vertices can force u , and S' is a zero forcing set of W .

Case 2: Assume u does force a vertex v in W . Since u is adjacent to two or more pendant vertices, there exists a vertex $p \neq v$ that is adjacent to u and is a pendant vertex. Since u can only force one vertex, it must be the case that $p \in S$. Then, p can force u , which can then force v . Thus, S' is a zero forcing set of W .

In both cases, $S' = S \setminus \{u\}$ is a zero forcing set of W . Since this process can be repeated for any $u \in U$, $S'' = S \setminus U$ is a zero forcing set of W . Note that

$$|S''| = |S| - |U| < Z(W) + |U| - |U| = Z(W),$$

which is a contradiction since one cannot find a zero forcing set of W smaller than $Z(W)$. Thus, $|S| = Z(W) + |U|$ and, by Corollary 2.5 (2), $Z(W(U) \cap H) = Z(W)|V(H)|$. $\square$

3. Bounds using forcing chains

In this section, we provide an upper bound on the zero forcing number of hierarchical products by utilizing forcing chains. Forcing chains indicate the order that vertices are forced for a given initial set of vertices. The following terminology was first defined for digraphs in [3] and analogous definitions for undirected graphs are given in [2]. A *chronological list of forces* is a record of the forcing actions in the order in which they are performed. For each vertex w there may be more than one vertex that can force w . For a particular list of forces, a *forcing chain*, F is a sequence of forces $F = (v_1, v_2, \dots, v_k)$ such that $v_i \rightarrow v_{i+1}$ for $i = 1, \dots, k-1$. A *maximal forcing chain* is a forcing chain that is not a proper subsequence of another forcing chain. Note that a forcing chain F can consist of a single vertex (v_1), and such a chain is maximal if $v_1 \in S$ and v_1 does not perform a force. Let $\mathcal{F}_G = \{F_1, F_2, \dots, F_{|\mathcal{S}|}\}$ be a set of maximal forcing chains associated with the zero forcing set S of G . Note that S may not be a minimal zero forcing set. Let the *terminus* of $\mathcal{F}_G$, denoted $\text{Term}(\mathcal{F}_G)$, be the set of vertices that do not perform a force in $\mathcal{F}_G$ [11]. If $v \in \text{Term}(\mathcal{F}_G)$, then we say v is a *terminal vertex*. It may be the case that $\text{Term}(\mathcal{F}_G) \cap S \neq \emptyset$; that is, there could be vertices in the zero forcing set that do not perform a force. Let the *middle* of $\mathcal{F}_G$, denoted $\text{Mid}(\mathcal{F}_G)$, be the set of vertices that perform a force in $\mathcal{F}_G$ but are not in S . That is, $\text{Mid}(\mathcal{F}_G) = V(G) \setminus (S \cup \text{Term}(\mathcal{F}_G))$. Note that in [2], it is proven that forcing chains are reversible and that $\text{Term}(\mathcal{F}_G)$ is also a zero forcing set of G .

Theorem 3.1. Let W and H be connected graphs with root set U . Let S_W be any zero forcing set of W such that $U \subseteq S_W$, and $\mathcal{F}_W$ is a set of maximal forcing chains associated with S_W . Define $U_1 = U \cap \text{Term}(\mathcal{F}_W)$. Let S_H be any zero forcing set of H , and $\mathcal{F}_H$ a set of maximal forcing chains associated with S_H with $M_H = |\text{Mid}(\mathcal{F}_H)|$. Then,

$$Z(W(U) \cap H) \leq |S_W||V(H)| - |V(H)||U| + |S_H||U| + (|U| - |U_1|)M_H.$$

In particular, if $|S_W| = Z(W)$ and $|S_H| = Z(H)$, then

$$Z(W(U) \cap H) \leq Z(W)|V(H)| - |V(H)||U| + Z(H)|U| + (|U| - |U_1|)M_H.$$

Proof. Let S_W be a zero forcing set of W with $U \subseteq S_W$, and let $\mathcal{F}_W$ be a set of maximal forcing chains associated with S_W . Similarly, let S_H be a zero forcing set of H , and let $\mathcal{F}_H$ be a set of maximal forcing chains associated with S_H . Define $S = S_1 \cup S_2 \cup S_3$ to be the set of vertices such that:

- $S_1 = \text{Term}(\mathcal{F}_W) \times S_H$: in each row corresponding to a vertex in S_H , the set of vertices corresponding to $\text{Term}(\mathcal{F}_W)$.
- $S_2 = (\text{Term}(\mathcal{F}_W) \setminus U_1) \times \text{Mid}(\mathcal{F}_H)$: in each row corresponding to a vertex in $\text{Mid}(\mathcal{F}_H)$, the set of vertices corresponding to $\text{Term}(\mathcal{F}_W) \setminus U_1$.
- $S_3 = (S_W \setminus U) \times (V(H) \setminus (S_H \cup \text{Mid}(\mathcal{F}_H)))$: for the remaining rows, the set of vertices corresponding to $S_W \setminus U$.

Note that $|\text{Term}(\mathcal{F}_W)| = |S_W|$. It will be shown that S is a zero forcing set for $W(U) \cap H$ by filling vertices using the following three steps.

STEP 1: Consider a row corresponding to a vertex in S_H . The vertices in this row that are filled correspond to vertices in $\text{Term}(\mathcal{F}_W)$. Recall $\text{Term}(\mathcal{F}_W)$ is a zero forcing set of W . The set U consists only of vertices in S_W , and any vertex in U that is also in $\text{Term}(\mathcal{F}_W)$ does not force any other vertex. Therefore, vertices in that row associated with U do not block the reverse forcing process horizontally along that row. Thus, the vertices in S_1 horizontally force every row corresponding to the vertices in S_H . Since every row corresponding to a vertex in S_H is completely filled, vertices corresponding to S_H in columns corresponding to U may force a vertex vertically in the order of the forcing chains of $\mathcal{F}_H$. We continue to force vertices vertically until vertices in a row corresponding to a vertex in $\text{Mid}(\mathcal{F}_H)$ are forced or every row has had vertices forced vertically.

STEP 2: If it is the case that $\text{Mid}(\mathcal{F}_H) = \emptyset$, then move to STEP 3. Otherwise, consider the rows corresponding to newly forced vertices that are also in $\text{Mid}(\mathcal{F}_H)$. In these rows, vertices corresponding to U_1 are filled, and due to S_2 , the remaining vertices corresponding to $\text{Term}(\mathcal{F}_W)$ are filled in those rows. Thus, as in STEP 1, since vertices corresponding to $\text{Term}(\mathcal{F}_W)$ are filled in these rows, each of these rows can be completely forced horizontally. This process horizontally forces every

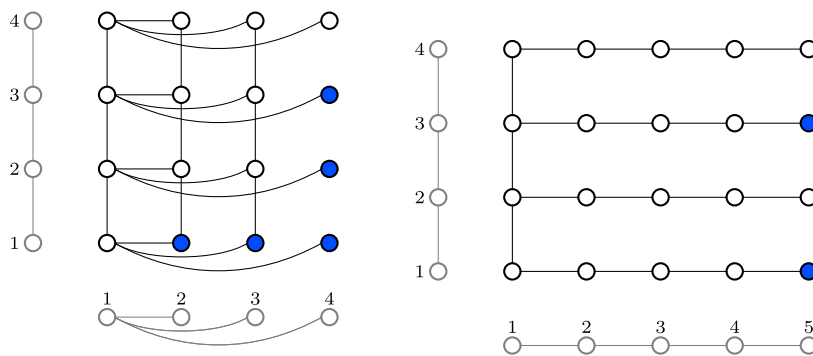


Fig. 3.1. On the left is $K_{1,3}(\{1, 2, 3\}) \sqcap P_4$ and on the right is $P_5(\{1\}) \sqcap P_4$.

row corresponding to a newly filled vertex that is also in $\text{Mid}(\mathcal{F}_H)$. In columns corresponding to U , the vertices in rows corresponding to a newly filled vertex that is also in $\text{Mid}(\mathcal{F}_H)$ may force vertically in the order of the forcing chains of $\mathcal{F}_H$. Continue to force as many vertices as possible. Now repeat STEP 2 until every row corresponding to a vertex in $\text{Mid}(\mathcal{F}_H)$ is filled completely.

STEP 3: After completion of STEP 2, the only rows that are not completely filled are those corresponding to $V(H) \setminus (S_H \cup \text{Mid}(\mathcal{F}_H))$. That is, the rows corresponding to vertices in $\text{Term}(\mathcal{F}_H)$, that are also not in S_H are not yet filled. Consider one of these rows. In this row, vertices corresponding to U are filled, and the remaining vertices corresponding to S_W are filled due to S_3 . Since all vertices in columns corresponding to U are filled in previous steps, any unfilled neighbor of a filled vertex is in the same row. Therefore, this row can be completely forced horizontally because vertices corresponding to S_W are filled in this row.

Thus, S is a zero forcing set of $W(U) \sqcap H$. Since $|\text{Term}(\mathcal{F}_W)| = |S_W|$,

$$\begin{aligned} |S| &= |S_1| + |S_2| + |S_3| \\ &= |\text{Term}(\mathcal{F}_W)| |S_H| + (|\text{Term}(\mathcal{F}_W)| - |U_1|) M_H \\ &\quad + (|S_W| - |U|)(|V(H)| - |S_H| - M_H) \\ &= |S_W| |S_H| + (|S_W| - |U_1|) M_H + (|S_W| - |U|)(|V(H)| - |S_H| - M_H) \\ &= |S_W| |V(H)| - |V(H)| |U| + |S_H| |U| + (|U| - |U_1|) M_H. \quad \square \end{aligned}$$

Note that Theorem 3.1 can be applied to any root set $U \subseteq V(W)$, and neither S_W nor S_H is required to be a minimum zero forcing set for W or H , respectively. In fact, in some cases, Theorem 3.1 provides a better upper bound when S_W or S_H is not a minimum zero forcing set.

Example 3.2. Consider the graph $K_{1,3}(\{1, 2, 3\}) \sqcap P_4$ shown on the left in Fig. 3.1. The set $S_W = U = \{1, 2, 3\}$ is not a minimum zero forcing for $K_{1,3}$. Using this S_W and $\mathcal{F}_W = \{1 \rightarrow 4, 2, 3\}$, then $\text{Term}_{\mathcal{F}_W} = \{2, 3, 4\}$, and $U_1 = \{2, 3\}$. If $S_H = \{1\}$, then $\mathcal{F}_H = \{1 \rightarrow 2 \rightarrow 3 \rightarrow 4\}$, $\text{Term}_{\mathcal{F}_H} = \{4\}$ and $\text{Mid}_{\mathcal{F}_H} = \{2, 3\}$. By Theorem 3.1, the set of blue vertices in Fig. 3.1 (on the left) is a zero forcing set and $Z(K_{1,3}(\{1, 2, 3\}) \sqcap P_4) \leq 5$. This bound is also better than any of the other known upper bounds.

Example 3.3. The graph $P_5(\{1\}) \sqcap P_4$ on the right of Fig. 3.1 uses a minimum zero forcing set for $W = P_5$, $S_W = U = \{1\}$. Using $\mathcal{F}_W = \{1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5\}$, $\text{Term}_{\mathcal{F}_W} = \{5\}$. However, the zero forcing set used for $H = P_4$ is $S_H = \{1, 3\}$. This is not a minimum zero forcing set for P_4 , but $\mathcal{F}_H = \{1 \rightarrow 2, 3 \rightarrow 4\}$ so that $\text{Mid}(\mathcal{F}_H) = \emptyset$. Then, $Z(P_5(\{1\}) \sqcap P_4) \leq 2$. If a minimum zero forcing set for P_4 is used, then the resulting bound on the zero forcing number is 3, which is not minimum for the product.

Corollary 3.4. Let W and H be any connected graphs. Let S_W be a minimal zero forcing set of W , and let $\mathcal{F}_W$ be a set of maximal forcing chains associated with S_W . Let S_H be a minimal zero forcing set of H , and let $\mathcal{F}_H$ be a set of maximal forcing chains associated with S_H . Let U be a root set such that U is a subset of the terminus of $\mathcal{F}_W$, and $U_1 = U \cap S_W$. If $U = U_1$ or $\text{Mid}(\mathcal{F}_H) = \emptyset$, then

$$Z(W(U) \sqcap H) = Z(W) |V(H)| - |V(H)| |U| + Z(H) |U|.$$

Proof. Assume $U = U_1$ or $\text{Mid}(\mathcal{F}_H) = \emptyset$. By Theorem 2.2,

$$Z(W) |V(H)| - |V(H)| |U| + Z(H) |U| \leq Z(W(U) \sqcap H).$$

By [Theorem 3.1](#), since $|U| - |U_1| = 0$ or $|Mid(\mathcal{F}_H)| = 0$,

$$Z(W(U) \cap H) \leq Z(W)|V(H)| - |V(H)||U| + Z(H)|U|. \quad \square$$

Example 3.5. Consider $W = C_n$, the cycle on n vertices. Let $U = \{u\}$, and let S_W contain u and one of its neighbors. Then, S_W is a minimum zero forcing set of C_n , and there exists a set of forcing chains for which u is a terminal vertex. Then, by [Corollary 3.4](#), $Z(C_n(\{u\}) \cap H) = |V(H)| + Z(H)$.

4. Graph constructions with large zero forcing number

In this section, the zero forcing number of hierarchical products constructed from graphs with large zero forcing numbers is given, including those products in which one factor is a complete graph K_n .

First, consider the case in which K_n is the factor of the hierarchical product that is associated with the root set.

Proposition 4.1. *Let H be any connected graph, and let $U \subseteq V(K_n)$ be a root set. If $|U| \leq n - 1$, then*

$$Z(K_n(U) \cap H) = (n - 1)|V(H)| - |V(H)||U| + Z(H)|U|.$$

If $|U| = n$, then $Z(K_n(U) \cap H) = Z(K_n \square H) \leq \min\{nZ(H), (n - 1)|V(H)|\}$.

Proof. Label the vertices of $K_n = \{1, \dots, n\}$. If $|U| = m \leq n - 1$, then without loss of generality, we can assume $U = \{1, \dots, m\}$. Let $S_{K_n} = \{1, 2, \dots, n - 2, n\}$ and $\mathcal{F}_{K_n} = \{1, 2, \dots, n - 2, n \rightarrow n - 1\}$. Recall $U_1 = U \cap \text{Term}(\mathcal{F}_{K_n})$. Then $U = U_1$ and $|S_{K_n}| = Z(K_n) = n - 1$. Let S_H be any minimum zero forcing set of H . By [Corollary 3.4](#),

$$Z(K_n(U) \cap H) = (n - 1)|V(H)| - |V(H)||U| + Z(H)|U|.$$

When $|U| = n$, we have the upper bound for the Cartesian product by [Lemma 2.4](#). $\square$

Next, consider when the complete graph is the factor of the hierarchical product not associated with the root set.

Proposition 4.2. *Let W be any connected graph, and let $U \subseteq V(W)$ be a root set. Let S_W be a zero forcing set of W such that $U \subseteq S_W$.*

- (1) *If S_W is a minimum zero forcing set of W , then $Z(W(U) \cap K_n) = nZ(W) - |U|$.*
- (2) *If S_W is not a minimum zero forcing set of W , then*

$$\max\{Z(W), n - 1, (n - 1)Z(W) - |U|\} \leq Z(W(U) \cap K_n) \leq n|S_W| - |U|.$$

Proof. Let S_W be a zero forcing set of W with an associated set of maximal forcing chains $\mathcal{F}_W$ such that U is a subset of the terminus of $\mathcal{F}_W$. Label the vertices of $K_n = \{1, \dots, n\}$. Let $S_{K_n} = \{1, 2, \dots, n - 1\}$ and $\mathcal{F}_{K_n} = \{1 \rightarrow n, 2, \dots, n - 1\}$. Note that $|S_{K_n}| = Z(K_n) = n - 1$ and $Mid(\mathcal{F}_{K_n}) = \emptyset$. By [Theorem 3.1](#),

$$Z(W(U) \cap K_n) \leq n|S_W| - n|U| + (n - 1)|U| = n|S_W| - |U|.$$

If $|S_W| = Z(W)$, we have equality by [Corollary 3.4](#). $\square$

Now consider other hierarchical products in which one factor has large zero forcing number.

Let $K_{1,t}$ be the star graph on $t + 1$ vertices. Note that $Z(K_{1,t}) = t - 1$.

Proposition 4.3. *Let H be any connected graph. Let $W = K_{1,t}$ with $t \geq 3$, where u is the vertex of degree t . Let $U \subseteq V(W)$ be a root set.*

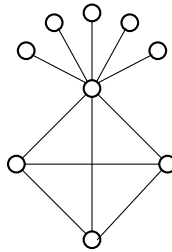
- (1) *If $|U| \leq t - 2$ and $u \notin U$, then $Z(W(U) \cap H) = Z(W)|V(H)| - |V(H)||U| + Z(H)|U|$.*
- (2) *If $U = \{u\}$, then $Z(W(U) \cap H) = Z(W)|V(H)|$.*

Proof. First, consider (1). Let S_W consist of U as well as $t - 1 - |U|$ additional pendant vertices of W . Note that $|U| < |S_W|$ since $|U| \leq t - 2$. Let $v \in S_W \setminus U$. Then, since it is a pendant vertex, v can force u , which in turn can force the last unfilled vertex in W . Thus, S_W is a zero forcing set of W , and it is minimum since $|S_W| = t - 1$. Let S_H be any minimum zero forcing set of H . By [Corollary 3.4](#), since $U = U_1$, $Z(W(U) \cap H) = Z(W)|V(H)| - |V(H)||U| + Z(H)|U|$.

Finally, (2) follows from [Corollary 2.5\(3\)](#). $\square$

Let $K_s \cup K_{1,t}$ be the pineapple graph on $s + t$ vertices for $s \geq 3$ and $t \geq 2$, which is constructed by appending t pendant vertices to a single vertex of the complete graph K_s . By [8], $Z(K_s \cup K_{1,t}) = s + t - 3$. (see [Fig. 4.1](#)).

Proposition 4.4. *Let H be any connected graph. Let $W = K_s \cup K_{1,t}$ with $s \geq 3$ and $t \geq 2$. Let $U \subseteq V(W)$ be a root set. Let u be the vertex of degree $s + t - 1$.*

Fig. 4.1. Pineapple graph $K_4 \cup K_{1,5}$.

(1) If U consists of

- at most $t - 2$ vertices of degree 1, and
- at most $s - 2$ vertices of degree $s - 1$,

then $Z(W(U) \cap H) = Z(W)|V(H)| - |V(H)||U| + Z(H)|U|$.

(2) If $U = \{u\}$, then $Z(W(U) \cap H) = Z(W)|V(H)|$.

Proof. First, consider (1). Let a_1 be the number of vertices of degree 1 in U , and a_{s-1} the number of vertices of degree $s - 1$ in U . Note that $|U| = a_1 + a_{s-1}$. Let S_W contain U as well as all other vertices of W , except for a vertex u of degree one, and a vertex of degree $s - 1$. We will show that S_W is a minimum zero forcing set of W such that $U = U_1$. Since U contains at most $t - 2$ vertices of degree 1, there exists $v \in S_W \setminus U$ such that $\deg(v) = 1$. Thus, v can force u . Also, since U contains at most $s - 2$ vertices of degree $s - 1$, there exists $w \in S_W \setminus U$ such that $\deg(w) = s - 1$. Then, w can force the last unfilled vertex of degree $s - 1$. Finally, u can force the last unfilled vertex of degree one. Thus, S_W is a zero forcing set of W and S_W is minimum since $|S_W| = s + t - 3$. Let S_H be any minimum zero forcing set of H . By Corollary 3.4, since $U = U_1$, $Z(W(U) \cap H) = Z(W)|V(H)| - |V(H)||U| + Z(H)|U|$.

Finally, (2) follows from Corollary 2.5. $\square$

Data availability

No data was used for the research described in the article.

References

- [1] AIM Minimum Rank - Special Graphs Work Group, F. Barioli, W. Barrett, S. Butler, S.M. Cioabă, D. Cvetković, S.M. Fallat, C. Godsil, W. Haemers, L. Hogben, R. Mikkelsen, S. Narayan, O. Pryporova, I. Sciriha, W. So, D. Stevanović, H. van der Holst, K. Vander Meulen, A. Wangsness, Zero forcing sets and the minimum rank of graphs, *Linear Algebra Appl.* 428 (2009) 1628–1648.
- [2] F. Barioli, W. Barrett, S. Fallat, H.T. Hall, L. Hogben, B. Shader, P. van den Driessche, H. van der Holst, Zero forcing parameters and minimum rank problems, *Linear Algebra Appl.* 433 (2010) 401–411.
- [3] F. Barioli, S.M. Fallat, D. Hershkowitz, H.T. Hall, L. Hogben, H. van der Holst, B. Shader, On the minimum rank of not necessarily symmetric matrices: a preliminary study, *Electron. J. Linear Algebra* 18 (2009) 126–145.
- [4] L. Barrière, F. Comellas, C. Dalfó, M.A. Fiol, The hierarchical product of graphs, *Discrete Appl. Math.* 157 (1) (2009) 36–48.
- [5] L. Barrière, C. Dalfó, M.A. Fiol, M. Mitjana, The generalized hierarchical product of graphs, *Discrete Math.* 309 (12) (2009) 3871–3881.
- [6] D. Burgarth, D. D'Alessandro, L. Hogben, S. Severini, M. Young, Zero forcing, linear and quantum controllability for systems evolving on networks, *IEEE Trans. Autom. Control* 58 (2013) 2349–2354.
- [7] D. Burgarth, V. Giovannetti, L. Hogben, S. Severini, M. Young, Logic circuits from zero forcing, *Nat. Comput.* 14 (2015) 485–490.
- [8] L. Dealba, J. Grout, L. Hogben, R. Mikkelsen, K. Rasmussen, Universally optimal matrices and field independence of the minimum rank of a graph, *Electron. J. Linear Algebra* 18 (2009) 403–419.
- [9] S. Fallat, L. Hogben, J.C.-H. Lin, B. Shader, The inverse eigenvalue problem of a graph, zero forcing, and related parameters, *Notices Amer. Math. Soc.* 67 (2020) 1–10.
- [10] S. Fallat, K. Meagher, B. Yang, On the complexity of the positive semidefinite zero forcing number, *Linear Algebra Appl.* 491 (2016) 101–122.
- [11] L. Hogben, M. Huynh, N. Kingsley, S. Meyer, S. Walker, M. Young, Propagation time for zero forcing on a graph, *Discrete Appl. Math.* 160 (2012) 1994–2005.
- [12] S. Jung, S. Kim, B. Kahng, Geometric fractal growth model for scale-free networks, *Phys. Rev. E* 65 (2002) 056101.
- [13] D. Kempe, J. Kleinberg, E. Tardos, Maximizing the spread of influence through a social network, in: *Proceedings of the Ninth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2003, 137146.
- [14] H. LeClair, T. Spilde, S.E. Anderson, B.K. Kroschel, Zero Forcing of Generalized Hierarchical Products of Graphs, Center for Applied Mathematics Student Research Paper, St. Paul, MN: University of St. Thomas, 2019, <http://dx.doi.org/10.48550/arXiv.2410.17440>.
- [15] M.E.J. Newman, The structure and function of complex networks, *SIAM Rev.* 45 (2003) 167–256.